

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1002

Idaho Springs 138 kV
August 30, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1002. The Interconnection Customer proposes a 13.6 MW generating facility interconnecting at the Idaho Springs 138 kV substation in the Virginia area. The request was received on May 25, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 386 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$55,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Centennial - Glen Carlyn 138 kV line	Idylwood - Gallows Rd 138 kV	119.4	202	6.7
Falls Church - Idylwood 138 kV	Centennial - Annandale 138 kV	111.1	200	7.7
Centennial - Annandale 138 kV	Gallows Rd - Idylwood Substation 138 kV	108.8	276	6.7
Idylwood - Gallows Rd 138 kV	Annandale - Centennial 138 kV	101.6	277	7.7
Gallows Road - Dessau 138 kV	Delamo - Grand Blanc Generating Sta 138 kV Line	100.5	478	12.6
South Houston - Desau 138 kV L	Del Amo - Grand Blanc Generating Sta 138 kV	100.1	493	12.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Centennial - Glen Carlyn 138 kV	Rebuild line with higher-capacity conductor	\$1,512,000	1% (\$15,000)
Falls Church - Idylwood 138 kV	Rebuild line with higher-capacity conductor	\$2,266,000	1% (\$24,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Centennial - Annandale 138 kV	Reconductor line	\$378,000	1% (\$4,000)
Gallows Rd - Idylwood Substation 138 kV	Reconductor line	\$465,000	1% (\$5,000)
Gallows Road - Dessau 138 kV	Reconductor line	\$587,000	0% (\$3,000)
South Houston - Desau 138 kV L	Reconductor line	\$838,000	0% (\$4,000)
	Total allocated to project		\$55,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.